Physical Significance and Governing Principles of the Shape Analytics Simulators

Course project, Mechanics of Solids

October 29, 2026

Contents

1 Purpose and structure **2**

2 Chapter 1: Mathematical preliminaries **2**

3 Chapter 2: Traction vector and stress tensor **3**

3.1 Traction . . . 3

3.2 Transformation . . . 3

4 Chapter 3: Stress equilibrium equations **3**

5 Chapter 4: Principal values, maximum shear and Mohr's circle **4**

6 Chapter 5: Strain tensor **5**

7 Chapter 6: Stress–strain relations **5**

8 Chapter 7: Axisymmetric problems **5**

8.1 Extension and torsion . . . 5

8.2 Inflation of a cylinder (Lamé) . . . 6

9 Chapter 8: Bending of beams **6**

10 Chapter 9: Theory of beams **7**

11 Chapter 10: Energy methods **8**

12 Chapter 11: Theory of failure **8**

13 The 3D stress field simulator **9**

14 Limitations **10**

15 Verification **10**

1 Purpose and structure

The Shape Analytics app is a suite of interactive solid mechanics simulators. Its aim is to make stress analysis *move*: the user changes a load, a geometry or a material and sees the stress state, the deformation and the safety margin respond immediately. Every simulator is tied to one or more chapters of the course notes (Chapters 1 to 11). This report states, chapter by chapter, the principle used, the equations implemented, and what the quantity means physically.

Each simulator provides the same features: a live diagram, analytic charts, an A/B comparison (a saved baseline is overlaid on every chart), and exports (CSV, SVG, PNG, a design specification, and a lab report). All inputs use N, mm and MPa unless the label says otherwise.

Simulator	Chapters	Main principle used
3D stress field	2, 3, 4, 5, 6	Cauchy's theorem, equilibrium, strain–displacement, Hooke's law, half-space (Boussinesq, Cerruti) solution
Mohr's circle	1, 2, 4	Tensor transformation, principal values, invariants, hydrostatic/deviatoric split
Beam analysis	3, 8, 9	Equilibrium, pure and non-uniform bending, Euler–Bernoulli and Timoshenko theory
Shaft torsion	7	Axisymmetric equilibrium and compatibility for twist
Axial loading	6, 7	Extension of a bar, superposition, thermoelastic strain
Pressure vessel	7	Inflation of a cylinder, Lamé solution
Thermal stress	6	Isotropic thermoelasticity
Failure theories	11	Maximum principal stress, strain, shear and distortional energy criteria
Energy methods	10	Strain energy, reciprocity, Castigliano's theorem
Column buckling	9	Euler buckling, effective length

Table 1: Simulators and the chapters they implement.

2 Chapter 1: Mathematical preliminaries

Stress and strain are second-order tensors. A tensor is a physical object that exists independent of coordinates; a matrix is its representation in one basis. For a rotation tensor $\boldsymbol{Q}$ the components change as $[\boldsymbol{T}]' = \boldsymbol{Q}[\boldsymbol{T}]\boldsymbol{Q}^T$, while scalar combinations such as the trace and determinant do not.

Physical significance. This is why Mohr's circle and the invariants are meaningful. The material does not care how an analyst chose the axes, so any failure rule must depend only on quantities that survive rotation.

In the app. The Mohr's circle simulator applies $\boldsymbol{Q}$ about z to the stress matrix and reports the rotated components next to the invariants.

3 Chapter 2: Traction vector and stress tensor

3.1 Traction

Cutting a body along a plane with outward unit normal $\boldsymbol{n}$ exposes a distributed force. The traction is the force per unit area,

$$\boldsymbol{t}(\boldsymbol{x};\boldsymbol{n}) = \lim_{\Delta A \to 0} \frac{\Delta \boldsymbol{F}}{\Delta A}. \tag{1}$$

Applying linear momentum balance to an infinitesimal tetrahedron, the body force and inertia terms scale with volume (h^3) and vanish faster than the surface terms (h^2). What remains is Cauchy's theorem,

$$\boldsymbol{t}_n = \boldsymbol{\sigma}\,\boldsymbol{n}, \qquad \boldsymbol{\sigma} = \sum_i \boldsymbol{t}_i \otimes \boldsymbol{e}_i. \tag{2}$$

Physical significance. Traction on every plane through a point is fixed by just three vectors (nine numbers). Normal traction $\sigma_n = \boldsymbol{n} \cdot \boldsymbol{\sigma}\boldsymbol{n}$ pulls or pushes the planes apart; the in-plane part τ slides them. Stress is therefore local force transmission, not a property of a particular plane.

3.2 Transformation

Under rotation of axes the stress matrix transforms as $[\boldsymbol{\sigma}]' = \boldsymbol{Q}[\boldsymbol{\sigma}]\boldsymbol{Q}^T$. For plane stress rotated by θ,

$$\sigma_{x'} = \tfrac{\sigma_x+\sigma_y}{2} + \tfrac{\sigma_x-\sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta, \tag{3}$$

$$\tau_{x'y'} = -\tfrac{\sigma_x-\sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta. \tag{4}$$

In the app. In the 3D stress field the applied traction on a painted area is decomposed into normal and shear parts using exactly $\sigma_n = \boldsymbol{t}\cdot\boldsymbol{n}$ and $\boldsymbol{t}_s = \boldsymbol{t} - \sigma_n\boldsymbol{n}$. The Mohr's circle simulator implements the transformation above.

4 Chapter 3: Stress equilibrium equations

Linear momentum balance for a small element gives

$$\nabla \cdot \boldsymbol{\sigma} + \rho\boldsymbol{b} = \rho\ddot{\boldsymbol{u}}, \tag{5}$$

and angular momentum balance, in the absence of body couples, gives $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ (so $\tau_{xy} = \tau_{yx}$ and so on). The traction boundary condition is $\boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}}$ on the loaded surface.

Physical significance. Equilibrium says that stress cannot be chosen freely: its gradient must balance the body force, which is why stress varies smoothly through a loaded body. Symmetry of $\boldsymbol{\sigma}$ means a pair of shear stresses always acts on perpendicular faces, so six independent components suffice.

In the app. The beam simulator uses the integrated form of equilibrium, $dV/dx = -w$ and $dM/dx = V$, to draw the shear-force and bending-moment diagrams. The mohr simulator accepts only symmetric shears for the same reason.

5 Chapter 4: Principal values, maximum shear and Mohr's circle

Principal planes are the planes on which traction is purely normal, so $\boldsymbol{\sigma n} = \lambda \boldsymbol{n}$. They solve the eigenvalue problem

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0, \tag{6}$$

with the invariants

$$I_1 = \operatorname{tr}\boldsymbol{\sigma}, \quad I_2 = \sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2, \quad I_3 = \det\boldsymbol{\sigma}. \tag{7}$$

The roots are real, the principal directions are orthogonal, and the principal stresses are the extreme values of the normal traction. The maximum shear stress is $\tau_{\max} = (\sigma_1 - \sigma_3)/2$ on planes at 45° to the extreme principal directions. In the (σ, τ) plane the state of every plane lies on or between three circles, with diameters $\sigma_1 - \sigma_3$, $\sigma_1 - \sigma_2$ and $\sigma_2 - \sigma_3$ (Mohr's circle and Mohr's stress plane). The stress splits into a hydrostatic part $p\boldsymbol{I}$ with $p = I_1/3$ (volume change) and a deviatoric part $\boldsymbol{s} = \boldsymbol{\sigma} - p\boldsymbol{I}$ (shape change).

Physical significance. Principal stresses are the largest and smallest tensile or compressive intensities a point feels, and they decide where a brittle material cracks. The maximum shear controls ductile slip. The hydrostatic part changes volume but not shape, and the deviatoric part changes shape; this separation reappears in the von Mises criterion.

In the app. The simulator solves the cubic with the trigonometric (Lode) form, draws the three circles, marks the rotating face as a point that moves twice as fast as the element, and reports I_1, I_2, I_3, p and the failure margin.

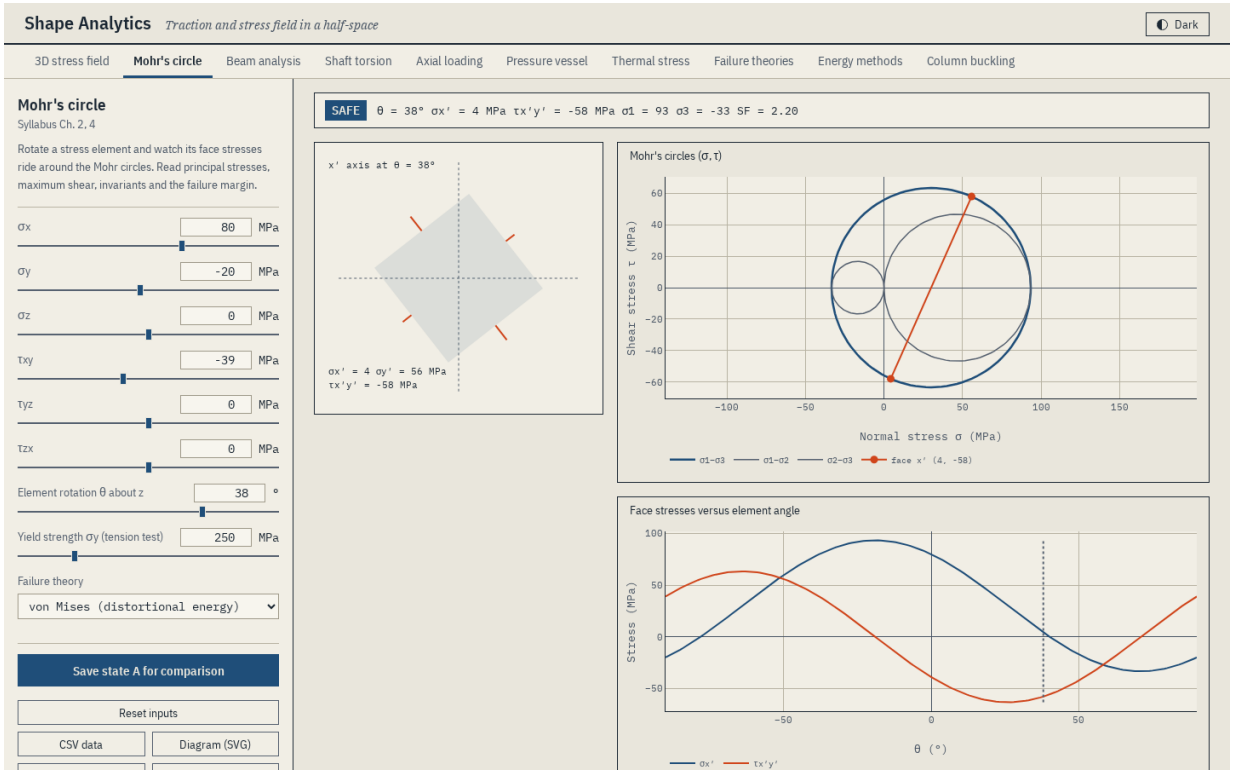


Figure 1: Mohr's circle simulator: the rotated element, the three circles and the face stresses versus angle.

6 Chapter 5: Strain tensor

Deformation is described by mapping a reference configuration to a deformed one. For small deformation the strain tensor is the symmetric part of the displacement gradient,

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right), \tag{8}$$

with longitudinal strain $\varepsilon_{xx} = \partial u_x/\partial x$, shear strain $\gamma_{xy} = 2\varepsilon_{xy}$, and volumetric strain $\varepsilon_v = \operatorname{tr}\boldsymbol{\varepsilon}$. The antisymmetric part is a local rigid rotation. Six strain components derive from three displacements, so they must satisfy compatibility conditions or the body would tear or overlap. Strain transforms exactly like stress, so there are principal strains and a Mohr's circle for strain.

Physical significance. Strain measures how much neighbouring points move relative to each other, after removing rigid motion. It is the quantity the material actually responds to, which is why constitutive laws link stress and strain.

In the app. The 3D stress field colours the body by strain magnitude and exaggerates the displacement field computed from the half-space solution so the deformed shape is visible. Axial, torsion and thermal simulators compute normal, shear and thermal strain respectively.

7 Chapter 6: Stress–strain relations

A linear elastic solid obeys $\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$. Symmetry of stress, strain and of the strain energy reduces the 81 constants to 21 for the most general material and to two for an isotropic one,

$$\sigma_{ij} = \lambda\,\varepsilon_{kk}\delta_{ij} + 2\mu\,\varepsilon_{ij}, \qquad \varepsilon_{ij} = \tfrac{1+\nu}{E}\sigma_{ij} - \tfrac{\nu}{E}\sigma_{kk}\delta_{ij}. \tag{9}$$

The engineering constants are related by $G = \mu = E/2(1+\nu)$ and $K = E/3(1-2\nu)$. With a temperature change the free strain $\alpha\Delta T\,\delta_{ij}$ is added, giving isotropic thermoelasticity $\varepsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\sigma_{kk}\delta_{ij} + \alpha\Delta T\delta_{ij}$.

Physical significance. E is stiffness in tension, G is stiffness in shear, ν is the sideways contraction that accompanies stretching, and K is resistance to volume change. Heating produces strain without stress; stress appears only if the supports prevent the strain.

In the app. The material selector fixes E, ν, α and σ_y for every simulator and the readouts show G and K. The thermal stress simulator implements the constrained cases

$$\sigma_{\text{bar}} = -E\alpha\Delta T, \quad \sigma_{\text{plate}} = \frac{-E\alpha\Delta T}{1-\nu}, \quad \sigma_{\text{block}} = \frac{-E\alpha\Delta T}{1-2\nu}. \tag{10}$$

The more directions are blocked, the stiffer the effective response and the larger the stress.

8 Chapter 7: Axisymmetric problems

In cylindrical coordinates (r, θ, z) the equilibrium equation in the radial direction reads $\frac{d\sigma_{rr}}{dr} + \frac{\sigma_{rr}-\sigma_{\theta\theta}}{r} = 0$ when the fields depend on r only. The chapter solves three canonical problems for a long cylinder: extension, torsion and inflation.

8.1 Extension and torsion

For extension, $\sigma_{zz} = N/A$ and $\delta = NL/EA$. For torsion, physical symmetry says plane sections stay plane and rotate rigidly, so $\gamma_{\theta z} = r\,d\phi/dz$ and, from Hooke's law,

$$\tau(r) = \frac{Tr}{J}, \qquad J = \frac{\pi}{32}(D^4 - d^4), \qquad \phi = \frac{TL}{GJ}. \tag{11}$$

Physical significance. Torque is carried only by shear stress that grows linearly with radius, so material near the axis is under-used and a hollow shaft keeps most of the stiffness at much lower mass. Torsional stiffness GJ/L plays the same role for twist that EA/L plays for stretch.

In the app. The axial simulator handles stepped bars in series, a fixed–fixed redundant case by compatibility $\sum \delta_i = 0$, and thermal strain. The torsion simulator plots $\tau(r)$ and $\phi(x)$ and compares the stiffness of the hollow and solid sections.

8.2 Inflation of a cylinder (Lamé)

Equilibrium plus Hooke's law plus compatibility give $\sigma_{rr} = A - B/r^2$ and $\sigma_{\theta\theta} = A + B/r^2$. The boundary conditions $\sigma_{rr}(a) = -p_i$ and $\sigma_{rr}(b) = -p_o$ fix

$$A = \frac{p_i a^2 - p_o b^2}{b^2 - a^2}, \qquad B = \frac{(p_i - p_o)a^2 b^2}{b^2 - a^2}, \tag{12}$$

and, for closed ends, $\sigma_{zz} = A$. For thin walls this reduces to $\sigma_\theta = pr/t$ and $\sigma_z = pr/2t$; for a sphere $\sigma_\theta = \sigma_\phi = pr/2t$.

Physical significance. Internal pressure tries to enlarge the vessel, and the wall resists with hoop tension, exactly as a hoop around a barrel does. The hoop stress is twice the axial stress, so a pressurised cylinder splits along its length. In a thick wall the hoop stress is largest at the inner surface, where the radial stress also equals $-p_i$.

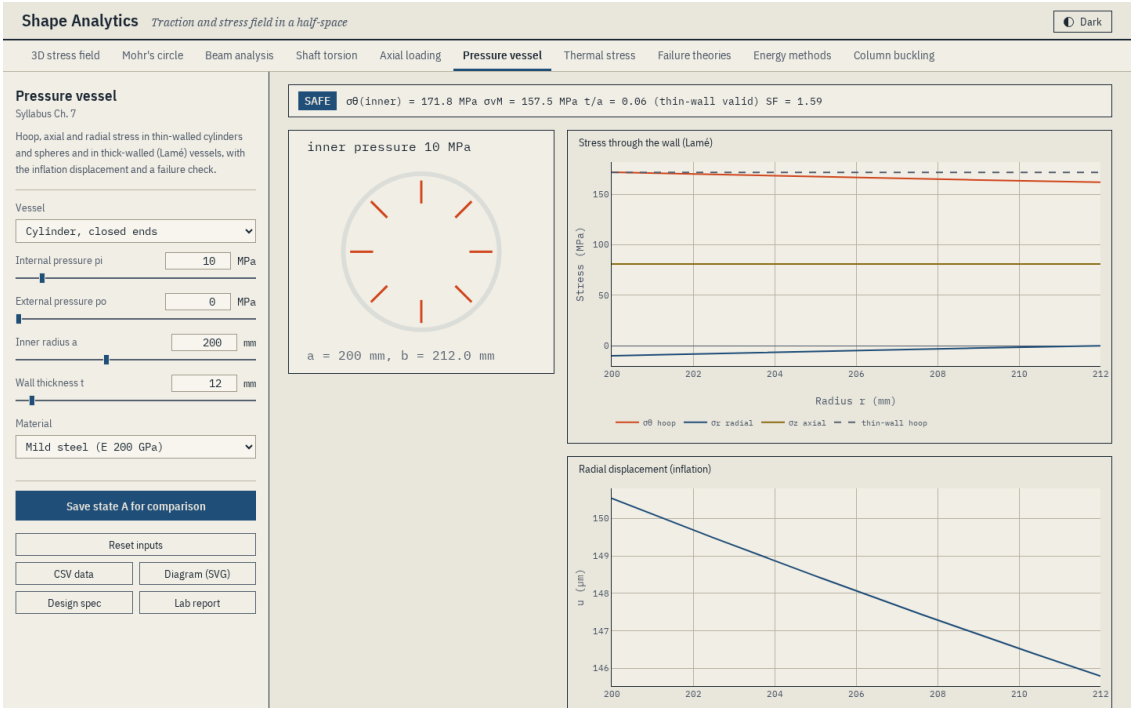


Figure 2: Pressure vessel simulator: Lamé stresses through the wall with the thin-wall hoop stress for comparison.

9 Chapter 8: Bending of beams

Pure bending. Plane sections remain plane and normal to the axis, so axial strain varies linearly with distance y from the neutral axis. With zero net axial force the neutral axis passes through

the centroid, and

$$\sigma_{xx} = -\frac{My}{I}, \qquad \frac{1}{\rho} = \frac{M}{EI}. \tag{13}$$

Non-uniform bending. When the moment varies, the shear force $V = dM/dx$ produces shear stress

$$\tau = \frac{VQ}{Ib}, \qquad Q = \int_y^c y'\, b(y')\, dy', \tag{14}$$

which is largest at the neutral axis (for a rectangle $1.5V/A$, for a solid circle $4V/3A$). Unsymmetrical sections need the product of inertia and a shear centre, the point through which the load must act to avoid twist.

Physical significance. The beam carries load by stretching one side and shortening the other, so fibres far from the neutral axis do the work, which is why deep or I-shaped sections are efficient. Shear is carried by the web, and the shear stress is zero at the outer surface.

In the app. The beam simulator solves statics for simply supported and cantilever beams under a point load, a uniform load and an applied moment, then integrates the curvature numerically. It plots V, M, deflection, and the through-depth distributions of σ and τ for rectangular, solid and hollow circular sections.

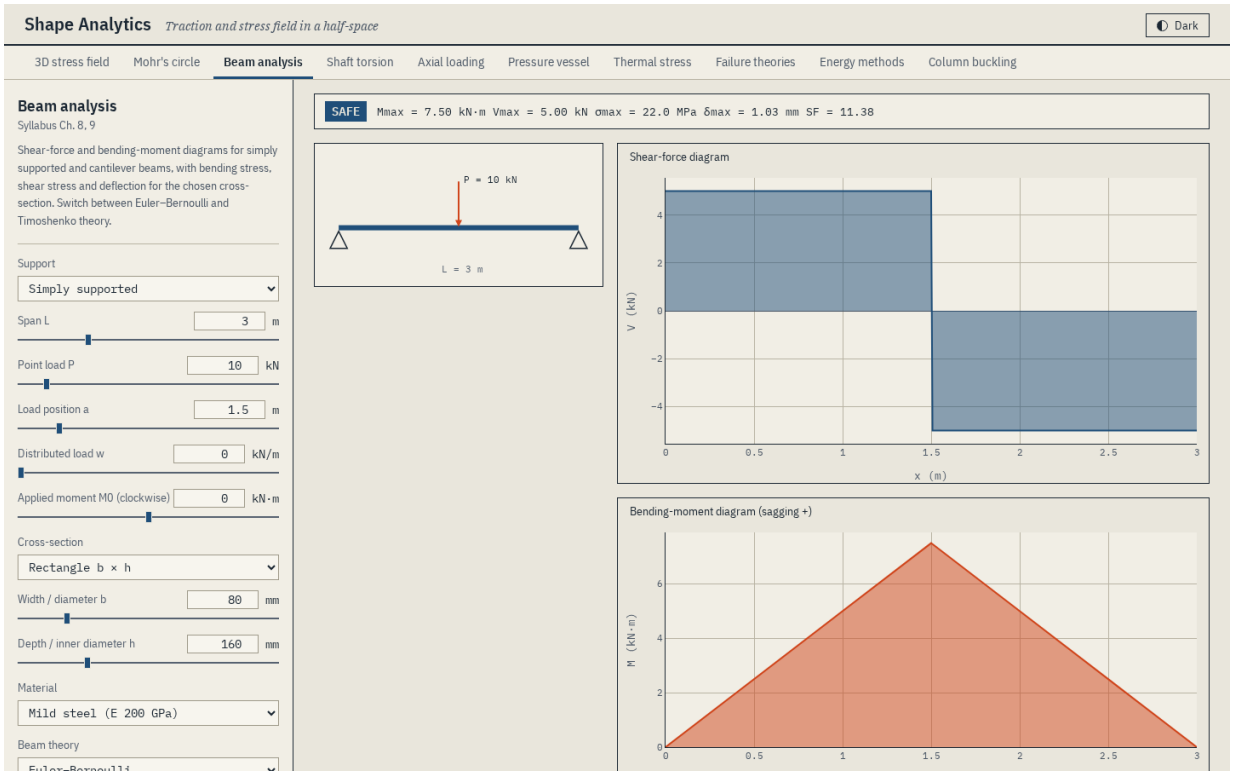


Figure 3: Beam simulator: loading diagram, shear force and bending moment.

10 Chapter 9: Theory of beams

Euler–Bernoulli theory assumes plane sections remain plane and normal to the deformed axis, giving $EI\, v'' = M(x)$. **Timoshenko theory** lets the section rotate independently of the axis, so shear deformation is retained. The extra deflection under shear is $\int V/(k_s GA)\, dx$, with shear correction factor k_s (for example $10(1+\nu)/(12+11\nu)$ for a rectangle). For a cantilever with a tip

load,

$$y_{\mathrm{TBT}} = \frac{PL^3}{3EI} + \frac{PL}{k_s GA}, \qquad \frac{y_{\mathrm{TBT}} - y_{\mathrm{EBT}}}{y_{\mathrm{TBT}}} = \frac{1}{1 + k_s GAL^2/3EI}, \tag{15}$$

so the error of Euler–Bernoulli theory grows as the beam becomes short and deep.

Buckling. A slender column under axial compression becomes unstable when

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}, \qquad \sigma_{cr} = \frac{\pi^2 E}{\lambda^2}, \qquad \lambda = \frac{KL}{r}, \tag{16}$$

with $K = 1,\ 0.7,\ 0.5,\ 2$ for pinned–pinned, fixed–pinned, fixed–fixed and fixed–free ends.

Physical significance. Buckling is a loss of stability, not of strength: the straight shape stops being the stable equilibrium at a load set by bending stiffness EI and geometry, often far below the yield load. Restraining the ends shortens the effective length and raises P_{cr} with the square of $1/K$.

In the app. The beam simulator has an Euler–Bernoulli / Timoshenko switch. The column buckling simulator plots σ_{cr} against slenderness with the yield cut-off and draws the first buckled mode shape for each end condition.

11 Chapter 10: Energy methods

Work done on a linear elastic body is stored as strain energy $U = \frac{1}{2}\int \boldsymbol{\sigma} : \boldsymbol{\varepsilon}\, dV = \frac{1}{2}\sum P_i \delta_i$. For common members

$$U_{\mathrm{axial}} = \frac{N^2 L}{2EA}, \quad U_{\mathrm{torsion}} = \frac{T^2 L}{2GJ}, \quad U_{\mathrm{bending}} = \int \frac{M^2}{2EI} dx, \quad U_{\mathrm{shear}} = \int \frac{V^2}{2k_s GA} dx. \tag{17}$$

Maxwell–Betti reciprocity: $P_a \delta_{ab} = P_b \delta_{ba}$. **Castigliano's first theorem:** the displacement at a load, in the direction of the load, is

$$\delta_i = \frac{\partial U}{\partial P_i}. \tag{18}$$

Physical significance. Because stored energy depends only on the final loads and not on how they were applied, one scalar function U contains all the load–displacement information. Reciprocity is a consequence of linearity: influence is symmetric.

In the app. The energy simulator computes U for four members, differentiates $U(P)$ numerically to obtain δ, and compares it with the closed form. It also shows reciprocity on a simply supported beam, where the deflection at $0.7L$ from a load at $0.3L$ equals the reverse. Switching shear energy on reproduces the Timoshenko result of Chapter 9.

12 Chapter 11: Theory of failure

A failure theory turns a stress state into one equivalent value that is compared with the critical value σ_{cr} of a simple tension test. The criteria implemented are:

Theory	Failure condition	Best for
Maximum principal stress	$\max \lvert\sigma_i\rvert \geq \sigma_{cr}$	brittle fracture
Maximum normal strain	$\varepsilon_1 = [\sigma_1 - \nu(\sigma_2 + \sigma_3)]/E \geq \sigma_{cr}/E$	brittle materials
Maximum shear (Tresca)	$\sigma_1 - \sigma_3 \geq \sigma_{cr}$	ductile metals (conservative)
Distortional energy (von Mises)	$\sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \geq \sigma_{cr}$	ductile metals

The octahedral shear stress is $\tau_{oct} = \frac{\sqrt{2}}{3}\sigma_{vM}$, so the octahedral shear and distortional energy theories predict the same envelope.

Physical significance. Ductile metals yield by shear, which depends only on the deviatoric part of stress: a hydrostatic pressure, however large, does not yield them. This is why Tresca and von Mises depend only on differences of principal stresses. Plotted in the σ_1–σ_2 plane, the von Mises criterion is an ellipse, Tresca a hexagon inscribed in it, and the maximum principal stress theory a square.

In the app. Every simulator that produces a stress state also reports a safety factor σ_y/σ_{eq}. The failure simulator draws all envelopes together with the operating point and reports the governing theory.

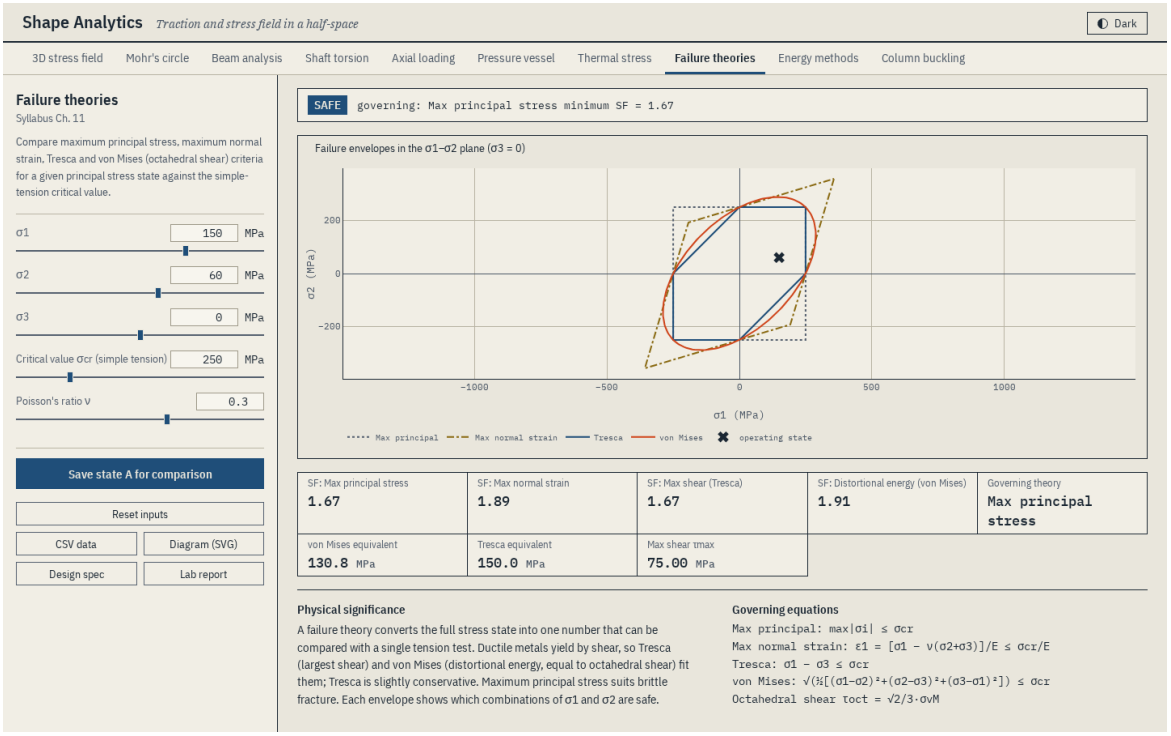


Figure 4: Failure theory simulator: envelopes in the σ_1–σ_2 plane and safety factors.

13 The 3D stress field simulator

The original module of the app combines Chapters 2 to 6 for a body loaded by a traction on an arbitrary painted area. The stress inside a half-space under a point force is given by the

Boussinesq solution for a normal load and the Cerruti solution for a tangential load; both satisfy equilibrium, compatibility and the traction boundary condition. A distributed load is obtained by superposition over the painted area, which is valid because the governing equations are linear (Chapter 10, linearity and superposition). The stress tensor at each point is converted to σ_n, $\tau_{\max}$, the von Mises stress and strain magnitude using Chapters 4 to 6, and the displacement field gives the deformed shape.

Physical significance. The model shows how a local load spreads with depth: stress decays rapidly away from the loaded area (St. Venant's principle) while the volume of loaded material grows. It also demonstrates that a purely normal surface traction produces shear stress below the surface, which is where yielding can start.

14 Limitations

- All models are linear elastic and small-strain. Plasticity (Chapter 6.8), large rotations and dynamics are not modelled.
- Beam deflections are obtained by numerical integration on 401 points; the Timoshenko correction uses a constant shear factor.
- The Lamé solution assumes a long vessel with closed ends and uniform temperature. The half-space solution is used as an approximation for a finite body in the 3D field.
- The column simulator assumes an ideal straight, concentrically loaded column; imperfections lower the real buckling load.

15 Verification

The closed-form results of each simulator were checked by hand for the default case: a simply supported beam with a central load gives $M_{\max} = PL/4 = 7.5$ kN m and $\delta = PL^3/48EI = 1.03$ mm; the circular shaft gives $\tau_{\max} = TR/J = 32.6$ MPa and $\phi = 1.17°$; the plane-stress Mohr circle gives $\sigma_{1,3} = 93.4, -33.4$ MPa; the Lamé hoop stress at the inner wall agrees with pr/t to 0.1% for $t/a = 0.06$; and Castigliano's derivative of U reproduces $PL^3/3EI$ for the cantilever.